

MIDTERM I

ECON498–ADVANCED TOPICS IN ECONOMICS: GENERAL EQUILIBRIUM

APRIL 5, 2026

1. (10 points each) Prove or disprove the following two statements.

(a) For any two sets S and T , $\text{co}(S \cup T) = \text{co}(S) \cup \text{co}(T)$.

(b) If S is convex, then $\bigcup_{x \in S} B_\varepsilon(x)$ is convex for every $\varepsilon > 0$.

2. A set $S \subseteq \mathbb{R}^n$ is a cone if $\lambda s \in S$ whenever $s \in S$ and $\lambda \in [0, \infty)$. For every set S , define the set

$$S^* = \{ p : p \cdot s \leq 0 \text{ for every } s \in S \}.$$

It is easily observed that for every S , S^* is a closed and convex cone. Accept the following two statements as correct.

cat If S is a cone and if there exist a vector p and a scalar c such that $p \cdot s < c$ for every $s \in S$, then $c > 0$ and $p \cdot s \leq 0$ for every $s \in S$.

raccoon If S is a closed and convex cone, then $(S^*)^* = S$.

Consider the following statement.

Let S be a closed and convex cone and K be a compact and convex set. Then

$$K \cap S^* \neq \emptyset \iff \forall s \in S \exists k \in K : s \cdot k \leq 0.$$

(a) (15 points) Prove this result using statements cat and raccoon.

(b) (5 points) Prove statement cat.

(c) (10 points) Prove statement raccoon using statement cat.

3. (20 points) Let $X \subseteq \mathbb{R}_+^m$ be nonempty and compact, and fix a vector $e \in X$. Interpret X as the consumption set and e as the endowment of a consumer. Define the consumer's budget correspondence¹ $\beta : \mathbb{R}_{++}^m \rightrightarrows X$ by

$$\beta(p) = \{ x \in X : p \cdot x \leq p \cdot e \}$$

for every $p \in \mathbb{R}_{++}^m$. Prove or disprove the following statement.

The budget correspondence is lower hemicontinuous.

4. (20 points) The following result is known as the **Intermediate Value Theorem** or **Bolzano's Theorem**, which hopefully you learned in Calculus I.

If $f : [a, b] \rightarrow \mathbb{R}$ is continuous, and if y lies between $f(a)$ and $f(b)$, i.e., if $f(a) \leq y \leq f(b)$ or $f(b) \leq y \leq f(a)$, then there exists some $x \in [a, b]$ such that $f(x) = y$.

Prove that the Intermediate Value Theorem implies **Brouwer's Fixed Point Theorem** when $m = 1$. (If $K \subseteq \mathbb{R}^m$ is nonempty, compact and convex and if $f : K \rightarrow K$ is continuous, then f has a fixed point.)

5. Let $K \subseteq \mathbb{R}^\ell$ be a nonempty closed set. The *support function* of K is the function $\mu_K : \mathbb{R}^\ell \rightarrow \overline{\mathbb{R}}$ defined² by

$$\mu_K(p) = \inf_{x \in K} p \cdot x.$$

For each $p \in \mathbb{R}^\ell \setminus \{0\}$ and $\alpha \in \mathbb{R}$, write

$$H^+(p, \alpha) = \{ x \in \mathbb{R}^\ell : p \cdot x \geq \alpha \}$$

for the closed half-space induced by normal vector p and scalar α . We also set $H^+(0, \alpha) = \mathbb{R}^\ell$ if $\alpha \leq 0$ and $H^+(0, \alpha) = \emptyset$ if $\alpha > 0$.

Fact (accept without proof). Let $f : \mathbb{R}^\ell \rightarrow \overline{\mathbb{R}}$ be concave. A vector $v \in \mathbb{R}^\ell$ is a *supergradient* of f at $\bar{p}$ if

$$f(p) \leq f(\bar{p}) + v \cdot (p - \bar{p}) \quad \text{for all } p \in \mathbb{R}^\ell.$$

The *subdifferential* of f at $\bar{p}$ is the set of all supergradients, denoted $\partial f(\bar{p})$. Then f is differentiable at $\bar{p}$ if and only if $\partial f(\bar{p})$ is a singleton $\{v\}$, in which case $\nabla f(\bar{p}) = v$.

¹Recall that $\mathbb{R}_+ = [0, \infty)$ and $\mathbb{R}_{++} = (0, \infty)$.

²Recall that $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$.

- (a) (5 points) Show that μ_K is **concave** and **homogeneous of degree one**, i.e. $\mu_K(\lambda p) = \lambda \mu_K(p)$ for all $\lambda \geq 0$.
 (b) (10 points) Suppose K is convex. Show that

$$K = \bigcap_{p \in \mathbb{R}^\ell} H^+(p, \mu_K(p)).$$

Hint: one inclusion is immediate from the definition of μ_K . For the other, use a Separation Theorem.

- (c) (5 points) Show that

$$\partial \mu_K(\bar{p}) = \{\bar{x} \in K : \bar{p} \cdot \bar{x} = \mu_K(\bar{p})\}.$$

That is, the subdifferential of μ_K at $\bar{p}$ is exactly the set of minimizers of $p \mapsto p \cdot x$ over K , evaluated at $\bar{p}$.

- (d) (5 points) Using parts (a)–(c) and the Fact above, prove the following.

Theorem (The Duality Theorem). Let $K \subseteq \mathbb{R}^\ell$ be a nonempty closed set and let μ_K be its support function. Then there exists a unique $\bar{x} \in K$ such that $\bar{p} \cdot \bar{x} = \mu_K(\bar{p})$ if and only if μ_K is differentiable at $\bar{p}$. Moreover, in this case,

$$\nabla \mu_K(\bar{p}) = \bar{x}.$$

- (e) (15 points) Let $K \subseteq \mathbb{R}^\ell$ now be an arbitrary nonempty closed set (not necessarily convex). Show that

$$\bigcap_{p \in \mathbb{R}^\ell} H^+(p, \mu_K(p)) = \overline{\text{co}(K)}.$$

Conclude that $\mu_K = \mu_{\overline{\text{co}(K)}}$, i.e. the support function cannot distinguish K from its closed convex hull.

- (f) (20 points) Let $f : \mathbb{R}_+^\ell \rightarrow \mathbb{R}_+$ be a production function that is **strictly quasiconcave** and **continuous**. Given an output level $q > 0$ and input prices $w \in \mathbb{R}_{++}^\ell$, the *cost function* and *conditional input demand* are defined by

$$c(w, q) = \min_{x \geq 0} \{w \cdot x : f(x) \geq q\}, \quad x(w, q) = \arg \min_{x \geq 0} \{w \cdot x : f(x) \geq q\}.$$

Use the Duality Theorem to prove **Shephard's Lemma**:

$$\nabla_w c(w, q) = x(w, q).$$

Hint: identify $c(w, q)$ as the support function of an appropriate set, and argue that $x(w, q)$ is unique.

6. (10 points) Below is a picture of Panqueque, General Equilibrium aficionado, Dragonborn,³ and Cat-Emperor of the Universe.



Panqueque being mysterious, circa Winter 2025.

Panqueque has provided emotional support in constructing the lecture notes, from his base of operations in Mexico. Based on your knowledge of Panqueque and on his mysterious stance above, what is Panqueque's **favorite result or theorem** covered in this course? **Justify your answer.**

Note: answers such as "he has no favorite theorem, he is a cat" will receive zero points. Panqueque is a connoisseur and will be grading this question himself.

³A being with the soul of a dragon.